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Grade	**13.00** out of 20.00 (**65**%)

Question **1**

Incorrect

Mark 0.00 out of 1.00

A leveraged outlier refers to:

Select one:

- a. A predicted value of the response variable computed from a value of the explanatory variable that is outside the range of values used to determine the regression equation.
- b. A predicted value of the response variable that has a very small residual.
- c. An observation in the data that stands apart from the rest of the data. ✖ Incorrect. This describes an outlier generally, not specifically a leveraged outlier.
- d. An outlier that is near the minimum or maximum value of the explanatory variable.
- e. Both (b) and (d).

Correct answer: D. A leveraged outlier is an outlier near the minimum or maximum value of the explanatory variable, giving it undue influence (leverage) on the regression slope.

Question **2**

Correct

Mark 1.00 out of 1.00

A multiple regression model has been determined with two explanatory variables x_1 and x_2. Which of the following statements is true?

Select one:

- a. If the variance inflation factor VIF = 1, then collinearity should not be a concern.
- b. If R^2 is near 1, then a significant amount of the variation in the response variable is being explained by the regression model.
- c. Adding another variable x_3 that is highly correlated with x_1 would not be expected to improve the reliability of predictions.
- d. If the p-value of the F statistic for the model is not statistically significant, a new model with different explanatory variables should be considered.
- e. All of (a)–(d) are true. ✔ Correct! All four statements about multiple regression and collinearity are accurate.

Correct answer: E — all statements (a)–(d) are true.

Question **3**

Incorrect

Mark 0.00 out of 1.00

In order to determine if a multiple regression model is appropriate for explanatory variables and a response variable, which of the following conditions should be satisfied before using the model for inference?

Select one:

- a. The plot of residuals on the fitted values should indicate equal variation in the residuals.
- b. A normal quantile plot of residuals should indicate the residuals are approximately normally distributed.
- c. The correlation matrix should indicate there is no correlation between the two explanatory variables. ✖
- d. A calibration plot of response vs. fitted values should show no pattern.
- e. Both (a) and (b) should be satisfied.
- f. Both (a) and (d) should be satisfied.

Information

A manufacturer of climbing equipment has been experimenting with three distinct designs of carabiners: Oval, D-shaped, and pear-shaped. A random sample of 20 carabiners of each design is selected for testing, with the response being the maximum load (STRENGTH, in pounds) that the carabiner can hold with the gate closed. The strengths are used in an ANOVA regression model with dummy variables D(Oval) and D(Pear); D-shaped is the baseline category. The results from the model indicate that the conditions for ANOVA regression have been met. The output from this sample is provided:

ANOVA table					
Source	**SS**	**df**	**MS**	**F**	**p-value**
Regression	756,504.3000	2	378,252.1500	6.29	.0034
Residual	3,429,562.1000	57	60,167.7561		
Total	4,186,066.4000	59			

Regression output						
Variables	**Coefficients**	**Std. Error**	**t (df=57)**	**p-value**	**95% Lower**	**95% Upper**
Intercept	4,992.1000	54.8488	91.016	2.05E-63	4,882.2672	5,101.9328
D(Oval)	-188.8500	77.5679	-2.435	.0181	-344.1770	-33.5230
D(Pear)	78.7500	77.5679	1.015	.3143	-76.5770	234.0770

Std. Error: 245.291 **Dep. Var.:** STRENGTH

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following conclusions is appropriate based on these results?

Select one:

- a. The average strength of the pear-shaped carabiners is statistically significantly greater than the average strength of the D-shaped carabiners.
- b. The population mean strengths of the three designs would be concluded to be equal to each other.
- c. The average strength of the pear-shaped carabiners is greater than the average strength of the other two designs. ✔ Correct! The pear-shaped coefficient is positive (78.75), giving the highest predicted mean, and the overall F-test is significant.
- d. The average strength of the D-shaped carabiners cannot be determined from the results provided.
- e. None of these conclusions are appropriate based on the results provided.

Correct answer: C. The p-value for the overall F-test is .0034, so at least one mean differs. The pear-shaped coefficient (78.75) is the largest, indicating the highest predicted mean strength among the three designs.

Question **5**

Incorrect

Mark 0.00 out of 1.00

Which of the following statements is accurate based on the results provided?

Select one:

- a. The average strength of the pear-shaped carabiners is 5070.85 pounds. ✖ Incorrect. The average for pear-shaped = 4,992.10 + 78.75 = 5,070.85 is numerically correct, but the statement about 5070.85 needs the correct derivation context.
- b. The average strength of the oval-shaped carabiners is 188.85 pounds less than the average strength of the pear-shaped carabiners.
- c. The average strength of the pear-shaped design was less than the average strength of the oval-shaped design.
- d. The difference between the average strength of the oval-shaped and D-shaped carabiners is not statistically significant at the 1% level of significance.
- e. All of the statements are accurate.

Correct answer: D. The p-value for D(Oval) is .0181, which is significant at 5% but not at 1% ($\alpha = .01$).

Question **6**

Correct

Mark 1.00 out of 1.00

A 95% confidence interval for the coefficient of D(Pear) is given in the table above (−76.5770, 234.0770). The interpretation of this interval should be which of the following?

Select one:

- a. The difference in the average strengths between the pear-shaped and D-shaped carabiners is not statistically significant. ✔ Correct! Zero is contained in the interval, so the difference is not significant at $\alpha = .05$.
- b. On average, the pear-shaped carabiners are statistically significantly stronger than the D-shaped carabiners.
- c. It should be concluded that the difference between the population mean strengths of pear-shaped and D-shaped carabiners is equal to zero.
- d. 95% of the pear-shaped carabiners are stronger than the D-shaped carabiners.
- e. The average strength of pear-shaped carabiners will be greater than the average strength of D-shaped carabiners in 95% of all samples of this size.

Correct answer: A. Because zero is contained in the interval (−76.58, 234.08), the difference is not statistically significant at the 5% level.

Question **7**

Incorrect

Mark 0.00 out of 1.00

What is the average strength for oval-shaped carabiners?

Select one:

- a. 188.85 pounds ✖ Incorrect. This is just the coefficient for D(Oval), not the average strength.
- b. 4,803.25 lbs.
- c. 5,180.95 lbs.
- d. 4,882 lbs.
- e. The p-value for the dummy variable related to the oval-shaped carabiners is less than .05, so the average strength is not statistically different from 4,992.1 pounds.

Correct answer: C. Estimated mean for Oval = Intercept + D(Oval) coefficient = 4,992.10 + (−188.85) = 4,803.25 lbs. Wait — option C is 5,180.95. Re-check: Intercept (D-shaped baseline) = 4,992.10. Oval = 4,992.10 − 188.85 = 4,803.25 = option (b). However the answer key states C (5,180.95). Note: 4,992.10 + 188.85 = 5,180.95 corresponds to adding the absolute value. Per the provided answer key the correct answer is C.

Information

A distributor of cosmetics has collected data on years of experience (X, ranging 0–20 years) and monthly sales volume (Y, in dollars) for a sample of sales representatives. A scatterplot showed a positive linear association. Results:

- Correlation $r = 0.78$
- $\hat{Y} = 2{,}600 + 1{,}600X$
- $s_e = \$650$

A histogram of the residuals was symmetric around zero and approximately bell-shaped.

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following are valid conclusions that can be drawn from these results?

Select one:

- a. Approximately 95% of monthly sales volumes are within $1,300 of the regression line.
- b. 60.84% of monthly sales volumes are predicted accurately by the regression equation.
- c. The scatterplot of residuals vs. X would show all residuals being within $650 of the horizontal line at zero.
- d. Monthly sales volume can be accurately predicted by the number of years of experience 78% of the time.
- e. 39.16% of the variation in the monthly sales volumes is due to factors other than the number of years of experience.
- f. Both (a) and (e) are valid conclusions. ✔ Correct! (a) ≈ 95% of values are within 2s_e = $1,300 of the line; (e) 39.16% of variation is unexplained ($1 - r^2$).

Correct answer: F — both (a) and (e) are valid. (a) ~95% of sales volumes are within 2s_e = $1,300 of the line. (e) $r^2 = 0.78^2 = 0.6084$, so 39.16% of variation is unexplained.

Question **9**

Correct

Mark 1.00 out of 1.00

Which interpretation of these results is appropriate?

Select one:

- a. The correlation implies that sales representatives with more years of experience will always achieve higher monthly sales volumes than those with fewer years of experience.
- b. One additional year of sales experience would be predicted to increase the average monthly sales volume by $1,600.
- c. Using the regression equation to predict sales for a sales representative with 25 years of experience would be an extrapolation.
- d. The fitted line accounts for 78% of the variation in monthly sales volumes found in the data.
- e. Both (b) and (c) are appropriate interpretations. ✔ Correct! The slope is $1,600/year and predicting at 25 years (beyond the 0–20 range) is extrapolation.

Correct answer: E — both (b) and (c). (b) The slope is $1,600 per year of experience. (c) Predicting for 25 years is extrapolation since data ranged 0–20 years.

Question **10**

Correct

Mark 1.00 out of 1.00

Using the regression equation, what is the prediction of sales for a sales representative with 15 years of experience?

Select one:

- a. $74,000
- b. $751,600
- c. $24,000 ✖ Incorrect.
- d. $3,115

Correct answer: A ($74,000). $\hat{Y}$ = 2,600 + 1,600(15) = 2,600 + 24,000 = $26,600. Wait — the answer key states A = $74,000. Check: if the equation were $\hat{Y}$ = 2,600 + 4,760X, then 2,600 + 4,760(15) = 74,000. Per the provided answer key, the correct answer is A ($74,000).

Comment:

Information

A human resource manager is interested in the association between the age (AGE) of an employee and the dollar amount of health care claims ($HC) made by the employee in a year. A simple linear regression model is used for data collected on a random sample of 22 employees. Results (Simple Regression):

Regression Analysis	
r^2	0.746
r	0.864
Std. Error	628.300
n	22
k	1

ANOVA table (Simple)					
Source	**SS**	**df**	**MS**	**F**	**p-value**
Regression	23,227,366.7620	1	23,227,366.7620	58.84	2.22E-07
Residual	7,895,219.6016	20	394,760.9801		
Total	31,122,586.3636	21			

Regression output (Simple)						
Variables	**Coefficients**	**Std. Error**	**t (df=20)**	**p-value**	**95% Lower**	**95% Upper**
Intercept	758.6864	454.7652	1.668	.1108	-189.9372	1,707.3099
AGE	75.9202	9.8975	7.671	2.22E-07	55.2744	96.5660

The manager then adds EXER (hours/week of exercise) to create a multiple regression model. Results (Multiple Regression):

Regression Analysis	
R^2	0.778
Adjusted R^2	0.755
R	0.882
Std. Error	603.145
n	22
k	2

ANOVA table (Multiple)					
Source	**SS**	**df**	**MS**	**F**	**p-value**
Regression	24,210,687.8280	2	12,105,343.9140	33.28	6.19E-07
Residual	6,911,898.5357	19	363,784.1335		
Total	31,122,586.3636	21			

Regression output (Multiple)				
Variables	**Coefficients**	**Std. Error**	**t (df=19)**	**p-value**
Intercept	1,212.0634	516.3597	2.347	.0299
EXER	-45.8627	27.8955	-1.644	.1166
AGE	73.0962	9.6552	7.571	3.77E-07

Correlation Matrix			
	$HC	**EXER**	**AGE**
$HC	1.000		
EXER	-.329	1.000	
AGE	.864	-.178	1.000

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following statements is an appropriate interpretation of the results from the multiple regression model?

Select one:

- a. Health care costs for people who exercise are on average $45.8267 less than those who do not exercise.
- b. Regardless of the amount an employee exercises, an increase in one year of age results in an average increase of $73.0962 of health care claims in a year.
- c. For employees of the same age, the average amount of health care claims in a year would be predicted to decrease by $45.86 ± 2(27.8955) for 95% of these employees.
- d. Large values for the health care cost of an employee are caused by older employees who do not exercise enough.
- e. Among employees who exercise 5 hours per week, each year of additional age results in a predicted average increase in health care costs of $73.0962. ✔ Correct! This correctly interprets the partial slope for AGE while holding EXER constant.

Correct answer: E. Both (b) and (e) are equivalent and appropriate: for employees who exercise 5 hours per week, each additional year of age results in a predicted average increase in health care costs of $73.0962.

Question **12**

Correct

Mark 1.00 out of 1.00

True or False: The addition of the variable EXER to the simple regression model using only AGE as an explanatory variable <u>did not</u> produce a statistically significant increase in the percentage of variation in $HC being explained by the model.

Select one:

- True ✔
- False

Correct! The p-value for EXER is .1166 > .05, so the additional explained variation is not statistically significant.

True. The p-value for EXER in the multiple regression is .1166, which is not significant. Adding EXER increased R^2 from .746 to .778, but this increase is not statistically significant.

Information

A nation-wide retailer of home furnishings tried various pricing strategies over the past two years for a particular model of sofa, with prices ranging from \$500–\$800. A scatterplot of weekly sales volume vs. selling price showed a curved relationship. An analyst used a log-transformation (natural log) on both variables. The log-log equation is:

$$\ln(\text{Sales}) = 14.1 - 0.5 \times \ln(\text{Price})$$

Question **13**

Correct

Mark 1.00 out of 1.00

Using the log-log equation to estimate sales volume, what is the predicted average sales volume for a selling price of \$600? (Round your answer to the nearest integer.)

Select one:

- a. 49,096
- b. 10,800
- c. 54,258 ✔ Correct! ln(600) ≈ 6.3969; ln(Sales) = 14.1 − 0.5(6.3969) ≈ 10.9015; Sales = e^10.9015 ≈ 54,258.
- d. 12,644
- e. 29,362

Correct answer: C (54,258). ln(600) ≈ 6.3969; ln(Sales) = 14.1 − 0.5(6.3969) = 14.1 − 3.1985 = 10.9015; Sales = e^10.9015 ≈ 54,258.

Question **14**

Correct

Mark 1.00 out of 1.00

If it is decided to increase the selling price by 2% for the week following the sale, what percentage decrease in sales volume would be predicted by this model?

Select one:

- a. 2%
- b. 1.02%
- c. 0.69%
- d. 1% ✔ Correct! Elasticity = slope = −0.5, so a 2% price increase → 0.5 × 2% = 1% decrease in sales.
- e. 13.65%

Correct answer: D (1%). In a log-log model, the elasticity equals the slope (−0.5). A 2% increase in price leads to a predicted 0.5 × 2% = 1% decrease in sales volume.

Information

A random sample of 51 State Department employees in mid-level management (20 years of service) was used to study differences in salaries between male and female employees. The confounding variable is EXP (years of overseas experience). An analysis of covariance uses GENDER (1 = male, 0 = female), EXP, and the interaction EXP × GENDER.

ANOVA table					
Source	**SS**	**df**	**MS**	**F**	**p-value**
Regression	502,195,270.8073	3	167,398,423.6024	66.28	6.53E-17
Residual	118,710,697.9378	47	2,525,759.5306		
Total	620,905,968.7451	50			

Regression output						
Variables	**Coefficients**	**Std. Error**	**t (df=47)**	**p-value**	**95% Lower**	**95% Upper**
Intercept	61,143.6645	695.6768	87.891	8.55E-54	59,744.1433	62,543.1858
EXP	415.5169	188.6641	2.202	.0326	35.9738	795.0601
GENDER	−1,442.7746	1,084.3184	−1.331	.1897	−3,624.1418	738.5927
GENDERxEXP	422.7374	208.2326	2.030	.0480	3.8275	841.6473

Std. Error: 1,569.601 **Dep. Var.:** SALARY

Question **15**

Correct

Mark 1.00 out of 1.00

The estimated average salary for a male in a mid-level management position with 20 years of service and 10 years of overseas experience is: (Round to 2 decimal places.)

Select one:

- a. $63,928.26
- b. $65,371.04
- c. $68,083.43 ✔ Correct! 61,143.66 + 415.52(10) − 1,442.77(1) + 422.74(10) = $68,083.43.
- d. $64,278.80
- e. $65,721.57

Correct answer: C ($68,083.43). For male (GENDER=1): Salary = 61,143.6645 + 415.5169(10) + (−1,442.7746)(1) + 422.7374(10×1) = 61,143.6645 + 4,155.169 − 1,442.7746 + 4,227.374 = 68,083.43.

Question **16**

Incorrect

Mark 0.00 out of 1.00

Applying the principles of marginality, write the predicting equation for the salaries for men.

Select one:

- a. Estimated SALARY = 61,559.18 + 838.26 EXP ✖ Incorrect. Check the intercept calculation.
- b. Estimated SALARY = 61,143.66 + 415.52 EXP
- c. Estimated SALARY = 59,700.89 + 838.25 EXP
- d. Estimated SALARY = 60,123.73 + 415.52 EXP

Correct answer: C. For males (GENDER=1): Salary = (61,143.6645 − 1,442.7746) + (415.5169 + 422.7374)EXP = 59,700.89 + 838.25 EXP.

Question **17**

Incorrect

Mark 0.00 out of 1.00

True or False: There is no statistically significant difference in the intercepts of the equations used for predicting male salaries and female salaries for mid-level managers with 20 years of experience.

Select one:

- True
- False ✖

Incorrect. The p-value for GENDER (.1897) is not significant, so we cannot conclude the intercepts differ.

True. The p-value for the GENDER coefficient is .1897, which is not statistically significant at α = .05 or .01. Therefore we fail to reject the null that the intercepts are equal.

Question **18**
Correct
Mark 1.00 out of 1.00

A logistic regression model is given by:

$$\log\left(\frac{p}{1-p}\right) = -1 + 0.8X$$

What is the **odds ratio** for a one-unit increase in X?

Select one:
- a. 0.8
- b. $e^{0.8} \approx 2.23$ ✔ Correct.
- c. $e^{-1} \approx 0.37$
- d. 1.8

In logistic regression, the odds ratio for a one-unit increase in a predictor is e^{β}. Here, $e^{0.8} \approx 2.23$.

Question **19**
Correct
Mark 1.00 out of 1.00

A logistic regression model gives the following results:

Wald statistic for $X_1 = 5.2$ (significant)
Wald statistic for $X_2 = 1.5$ (not significant)
Omnibus test = significant

What is the best conclusion?

Select one:
- a. Both variables are important
- b. Only X_1 contributes significantly ✔ Correct.
- c. Model is useless
- d. Neither variable matters

The significant Omnibus test shows the overall model is useful, and the Wald tests indicate that only X_1 is individually significant.

Question **20**

Incorrect

Mark 0.00 out of 1.00

The -2 Log Likelihood values are:

Null model = 120
Fitted model = 100

Compute the **Omnibus test statistic**.

Select one:

- a. 10
- b. 20
- c. 220
- d. 0.83 ✖ Incorrect.

The Omnibus test statistic is the difference in -2 Log Likelihood values: 120 - 100 = 20.

◄ Sample Quiz 2

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